

FORWARD DEPLOYED AI ENGINEERING

Capstone Project — CloudServe Solutions

Effort Log

How this log was kept. Entries were recorded as the work happened, in LEARNING_LOG.md and PROGRESS.md, both of which are committed to the repository with timestamps that predate this document. This log is assembled from those daily records rather than reconstructed from memory at the end, and the commit history is the evidence for that. Hours are actual hours worked, including the four on Tuesday 15 September that produced nothing assessable.

1 Details

Field	Value
Full name	Rajarajan Venkatesan
Project start date	6 September 2026
Target submission date	20 September 2026, 23:59
Coordinator review	15 September 2026 — five days before submission, by arrangement
Hours available per week	Roughly 21 to 26. The project runs alongside full-time work: weekday evenings at about 2.5 hours, weekend days at about 8
Total hours recorded	79.5 — of which 71.5 worked and 8.0 planned (video and packaging)

2 Summary of hours by stage

Completed last, from the daily entries below. The totals reconcile with those entries.

Stage	Planned	Actual	Difference	Why the difference
0. Orientation	1.0	2.0	+1.0	Reading the pack end to end and mapping the twelve acceptance criteria onto the four submission parts. Double the estimate and worth it — nothing later was built against a guess about what is graded
1. Discovery	6.0	8.0	+2.0	The analysis scripts took longer than the write-up. Cross-tabulating intent against answerability was not in the plan and is the single most valuable half-hour in the project
2. Requirements	3.0	3.0	0	On estimate. The only stage that landed exactly, and only because discovery had already done the thinking
3. Prompt library	2.5	2.5	0	On estimate
4. Sprint plan	1.5	1.5	0	On estimate
5. Build, gate and revision	31.5	44.5	+13.0	The largest overrun and the most informative. Four causes: the vector store had to be rewritten as pure Python (+2.0), routing had to be rebuilt as ordered gates (+4.0), the safety leaks needed diagnosis rather than a threshold change (+3.0), and model-provider access consumed 4.0 hours for nothing
6. Report, video, packaging	14.0	17.0	+3.0	The report ran to 8.0 against 6.0 estimated. Video (6.0) and packaging (2.0) are planned figures, not yet worked
Coordinator review	1.0	1.0	0	Held 15 September against a working system
Total	60.5	79.5	+19.0	A third over. Section 7 says where it went

3 Week one: 6 to 8 September

Discovery, requirements, prompts and planning, plus the first day of the build.

Date	Task	Stage	Hrs	What it produced	Blocked by anything?
Sat 6	Read the pack end to end; mapped the twelve acceptance criteria against the four submission parts	0	2.0	A one-page map of what is graded and where it is produced	No
Sun 7	Stakeholder interviews read; contradictions between accounts listed rather than resolved	1	2.5	Five accounts, three explicit disagreements to settle with data	No
Sun 7	Wrote the analysis scripts — answerability by intent, resolution time by segment, effort share vs volume share	1	3.0	The 70.7% and 47.5% figures; the finding that two intents are never answerable	No
Sun 7	Discovery workbook written up	1	2.5	Stage_1_Discovery_Workbook_Completed.docx	No
Mon 8	PRD v1.0 — 12 functional, 7 non-functional requirements, each traced to a finding	2	3.0	Stage_2_PRD_v1.docx	No
Mon 8	Prompt library — 7 versioned prompts with test notes	3	2.5	prompts/ directory, Stage_3 workbook	No
Mon 8	Sprint plan, two phases around the 15 Sept coordinator review	4	1.5	Stage_4_Sprint_Plan_Completed.docx	No
Tue 8	B-01 environment + repo scaffold; first live model call attempted	5	2.0	Repo structure, config module, CI skeleton	Yes — OpenRouter free tier returned 429 on the first call. Deferred; built the offline path instead
Tue 8	B-02 ingest and normalisation, four channels	5	1.5	src/ingest.py — never raises; customer name excluded from the object by design	No
Week 1	Week total		20.5	Stages 1-4 complete, build started	

4 Week two: 9 to 14 September

The build. The gate cleared on Monday 14 September, one day ahead of the coordinator review.

Date	Task	Stage	Hrs	What it produced	Blocked by anything?
Wed 9	B-02 finished; invisible-character and empty-body handling	5	2.5	Ingest tests green, 580/580 normalise with zero warnings	Yes — my own test file contained literal null bytes and would not parse. 30 min lost
Thu 10	B-03 chunking — section-aware vs fixed 800-char window, measured	5	2.5	87.7% vs 85.4% hit@1; section-aware chosen and the reason recorded	No
Fri 11	B-04 retrieval. Vector store abandoned; TF-IDF written in pure Python	5	2.5	src/retrieve.py, 145 passages, synonym expansion	Yes — chromadb and sentence-transformers would not install. Rewrote rather than fought it
Sat 12	Retrieval threshold sweep, 0.00 to 0.40 across 500 tickets	5	1.5	Precision caps at 78% at every value. Architectural finding, not a tuning result	No
Sat 12	B-06 classifier, 22 intents; B-07 routing; B-10 decision logging	5	6.5	End-to-end run on 20 tickets, every decision carrying a reason	No
Sun 13	B-08 generation with citations; B-09 both guardrail layers	5	4.0	src/generate.py, src/guardrails.py — five deterministic checks	No

Date	Task	Stage	Hrs	What it produced	Blocked by anything?
Sun 13	Diagnosed 23 safety leaks. Read all 23; found three classifier defects	5	3.0	Three detectors added, safety intents weighted; violations to zero without touching the threshold	No
Sun 13	B-05 harness computing real metrics	5	2.0	evaluation/harness.py with --input/--output	No
Mon 14	B-11 THE GATE — full 580-ticket unattended run	5	1.0	580/580, 0 errors, 2,490 decisions reconciling, 2.7 s	No
Mon 14	Induced provider outage; found and fixed the retry-storm	5	1.5	Circuit breaker. 20 tickets from 23 s back to seconds	No
Week 2	Week total		27.0	Gate cleared one day ahead of plan	

5 Week three: 15 to 20 September

Review, the Phase B documents, and packaging. The last two rows are planned rather than recorded, and are marked as such.

Date	Task	Stage	Hrs	What it produced	Blocked by anything?
Tue 15	B-18 test suite, B-19 clean-checkout verification, B-14 fairness segmentation	5	2.5	62 tests under one command; three fairness gaps found and kept	Yes — SQLite 'disk I/O error' on the second machine; added a temp-dir fallback
Tue 15	Coordinator review held against measured results	—	1.0	Discussion on merit rather than on intentions	No
Tue 15	Model provider access — Mistral key configured, tested, failed, re-tested	5	4.0	Nothing assessable. Account-level block, not a rate limit	Yes — this is the four hours that produced nothing. See section 7
Tue 15	Fixed the misleading 'no API key configured' error found during the above	5	0.5	has_key and has_http_library separated, each with its specific fix	No
Wed 16	B-16 PRD v2.0 and the Stage 5 revision log	5	2.5	4 requirements changed, 2 added; every entry names a measurement	No
Wed 16	B-17 governance framework — register, fairness audit, incident procedure	5	2.5	Governance_Framework_Completed.docx	No
Wed 16	Kill switch built, because writing the governance doc found it missing	5	1.5	src/kill_switch.py + 8 tests including fail-closed. Test count 62 → 78	No
Thu 17	B-15a report — five figures generated from the measured data; sections 1 to 6 drafted	6	4.0	Draft through Implementation; Figures 1 to 5	No
Fri 18	B-15a report — sections 7 to 10, five appendices, contents and cross-references	6	4.0	RajarajanVenkatesan_Capstone_Report.pdf, 30 pages	No
Fri 18	Effort log reconstructed from the daily learning log and progress tracker	6	1.0	This document	No
Fri 18	B-13 continuous integration verified running with no API key present	5	0.5	Workflow green on push	No
Sat 19	B-15b video — rehearsal and two takes	6	6.0	18–22 min, planned	Not yet done at time of writing
Sun 20	B-15c packaging and final checklist	6	2.0	Submission archive	Not yet done at time of writing
Week 3	Week total		32.0	Phase B documents complete; video and packaging remain	

6 Estimates against reality

The six largest items from the sprint plan.

Item	Est.	Actual	Why the difference
B-15b Video	6.0	6.0 (planned)	Not yet recorded. Two takes are planned, which is what the estimate assumed
B-15a Report	6.0	8.0	Underestimated by a third. Thirty pages of continuous prose with every claim traced to a measurement is slower than thirty pages of description. The figures were quick because the data was already computed
B-11 The gate	4.0	1.0	Four times faster than estimated, because the harness and the offline path were already right. The estimate assumed debugging during the run; there was none. This is the

Item	Est.	Actual	Why the difference
			clearest case of earlier work paying for later work
B-08 Generation	3.0	3.0	On estimate, because the prompt was written in Stage 3 before any code existed. The plan predicted this saving and it held
B-09 Guardrails	3.0	4.0	One hour over, spent replacing a length heuristic with claim-marker detection after finding that short replies do make claims
B-07 Routing	2.5	6.5	Two and a half times over, and the most valuable overrun in the project. The estimate covered deriving a threshold. What it actually took was discovering that a threshold cannot carry the decision, and rebuilding routing as four ordered gates
B-04 Retrieval	2.0	4.0	Double. The estimate assumed a vector store that installs. It did not, and TF-IDF with synonym expansion was written from scratch

7 Where the time went unexpectedly

Question	Answer
Which task took far longer than you expected, and why?	Routing, at 6.5 hours against 2.5. The estimate was for deriving a confidence threshold from the development data, which is an afternoon of analysis. What it became was the discovery that no threshold can separate answerable tickets from unanswerable ones — feature requests score higher than genuinely answerable tickets — followed by rebuilding routing as four ordered gates with safety ahead of confidence. The overrun was the project working correctly: the plan was wrong about the problem, and the extra four hours are what found that out. Diagnosing the twenty-three safety leaks inside that same stage cost another three hours and followed the same pattern.
Which task was easier than you expected, and why?	The gate run, at 1.0 hour against 4.0. The estimate assumed the usual pattern of starting an unattended run and debugging it when it stops halfway. It did not stop: 580 tickets, 2.7 seconds, zero errors, first attempt. That was bought earlier — the ingest layer that never raises, the harness built against the real input contract, and the decision log wired through every stage from the start. Four hours of estimate turned into one because the work had already been done somewhere else. The Stage 3 prompt library did the same thing for generation.
What would you allocate differently if you started again on Monday?	Two changes. First, build the evaluation harness before the components rather than alongside them. Every significant correction in this project came from a measurement — the chunking comparison, the threshold sweep, the three classifier defects, the circuit breaker — and each had to wait until something could be run over the full set and scored. Earlier measurement would have surfaced them sooner and more cheaply. Second, timebox infrastructure that earns no marks. Four hours went to model-provider access; a ninety-minute limit would have cost nothing and saved two and a half hours.
What did you spend time on that turned out not to matter?	Obtaining working model-provider access, at four hours on 15 September. The graded system was finished and passing at the time, and the pack is explicit that a well-built system on a small free model outscored a thin one on an expensive one. Each step had a plausible next hypothesis — pace the requests, widen the retry budget, correct the model identifier, switch provider — which is exactly how that kind of time disappears. The account was blocked at account level, not rate-limited, and no amount of client-side work was ever going to fix it. Two things were salvaged: the request-pacing and circuit-breaker code is genuinely useful, and the misleading error message found while debugging was a real defect that would have cost someone else the same evening.

8 Declaration

I confirm that this record is an accurate account of my own work on this project, assembled from daily records kept in the repository as the work happened.

Full name	Signature	Date	Total hours recorded
Rajarajan Venkatesan	R. Venkatesan	18 September 2026	79.5 (71.5 worked, 8.0 planned)